Annotated Bibliography

Bao, T., Klatt, B. N., Whitney, S. L., Sienko, K. H., & Wiens, J. (2019). Automatically evaluating balance: A machine learning approach. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, *27*(2), 179–186. https://doi.org/10.1109/TNSRE.2019.2891009

The purpose of this article was to determine whether machine learning algorithms can be implemented to grade home-based balance exercises. The authors observed that home-based balance training doesn't yield maximal results because often times there is no physical therapist present to give feedback during exercise. The authors collected trunk sway data from sixteen participants performing exercises. The physical therapist scored the participant's performance on exercises using a scale of 1-5. This data was used to implement machine learning by way of support vector machine modeling. The final model scored at an accuracy of 82% when compared to the physical therapist's scores. This was significantly higher than scores from the patients' self-assessments. This article displays the possibility for providing accurate monitoring of patient balance without direct therapist supervision.

Clark, R. A., Mentiplay, B. F., Pua, Y.-H., & Bower, K. J. (2018). Reliability and validity of the Wii Balance Board for assessment of standing balance: A systematic review. *Gait & Posture*, *61*, 40–54. https://doi.org/10.1016/j.gaitpost.2017.12.022

This paper was a review of 25 studies assessing the validity of a low cost Nintendo Wii Balance Board compared to valid laboratory equipment used for measuring standing balance. The authors explain how many researchers are starting to use this Wii Balance Board as it only costs approximately $100 and is extremely portable. The review found that overall the Wii Balance Board was a reliable and valid tool to measure balance. There were some caveats that scores can be affected by the duration of the test as well as what software is being used to collect the data. I found this study to be very helpful because it shows that we can use gaming equipment for clinical studies as long as it is validated.

Kang, S. H., Hwang, J., & Etebar Zadeh, M. (2026). A machine learning approach to predict objective and subjective low back fatigue using postural control features during sustained trunk flexion. *IISE Transactions on Occupational Ergonomics and Human Factors*. Advance online publication. https://doi.org/10.1080/24725838.2026.2646995

This study looked at the changes in posture when a person's lower back gets tired from bending forward for a long time. The researchers were trying to find a simple way to track this fatigue because traditional medical sensors are too annoying to wear normally when working a normal job. They had 18 people hold a bent posture and used machine learning models to predict how tired they were, both based on actual muscle data and how tired the participants felt. The machine learning models did a great job predicting the fatigue, especially an algorithm called Extra Trees. The study also found that wearing a supportive exosuit helped make the models more accurate. This research is important because it means companies could use smart shoes or insoles to tell when workers are getting tired, helping prevent back injuries.

Seo, J.-W., Kim, T., Kim, J. I., Jeong, Y., Jang, K.-M., Kim, J., & Do, J.-H. (2023). Development and application of a stability index estimation algorithm based on machine learning for elderly balance ability diagnosis in daily life. *Bioengineering*, *10*(8), 943. https://doi.org/10.3390/bioengineering10080943

The authors of this paper created a balance system designed to help elderly people check their balance during daily life. They built a flexible force mat that measures pressure and also used a Nintendo Wii Balance Board to test 103 older adults. By using machine learning, they created a formula that looks at how a person stands and accurately predicts their overall balance stability. When they compared their custom mat to the Wii Balance Board, they found out that the data matched up almost perfectly with a very high correlation rate.

Severini, G., Straudi, S., Pavarelli, C., Da Roit, M., Martinuzzi, C., Di Marco Pizzongolo, L., & Basaglia, N. (2017). Use of Nintendo Wii Balance Board for posturographic analysis of Multiple Sclerosis patients with minimal balance impairment. *Journal of NeuroEngineering and Rehabilitation*, *14*, Article 19. https://doi.org/10.1186/s12984-017-0230-5

This study focused on people with Multiple Sclerosis who are just starting to show very small balance issues. The scientists wanted to see if the Nintendo Wii Balance Board could spot these early balance problems just as well as an expensive laboratory force plate. They tested 18 patients and 18 healthy people by having them stand still with their eyes open and closed. Even though the Wii Balance Board slightly overestimated some numbers, both devices were over 80% accurate at telling the difference between a healthy person and someone with Multiple Sclerosis. The authors concluded that although the Wii might not give perfect numbers, it is still a great, low cost tool for doctors to compare different groups of patients and track how a disease is moving along.

Tapanya, W., & Sangkarit, N. (2024). Smartphone usage and postural stability in individuals with forward head posture: A Nintendo Wii Balance Board analysis. *Annals of Rehabilitation Medicine*, *48*(4), 289–300. https://doi.org/10.5535/arm.230034

This study investigated the impact of texting on a phone, on your balance. Especially for people who already have slightly forward head posture, which means their necks lean too far forward normally. The researchers gathered 53 young smartphone users, which probably wasn't very hard at all, and split them into two groups. One group had bad neck posture and one with normal posture. They made everyone stand on one leg and type on their phones while a Wii Balance Board measured how much they wobbled. The results showed that people who already had bad neck posture swayed around a lot more than the control group, especially when standing on a soft cushion. The authors concluded that using smartphones while walking or standing can impact your balance if you already have bad posture, which could put you at a higher risk for falls or hurting your neck.